High-Level Design (Detailed)

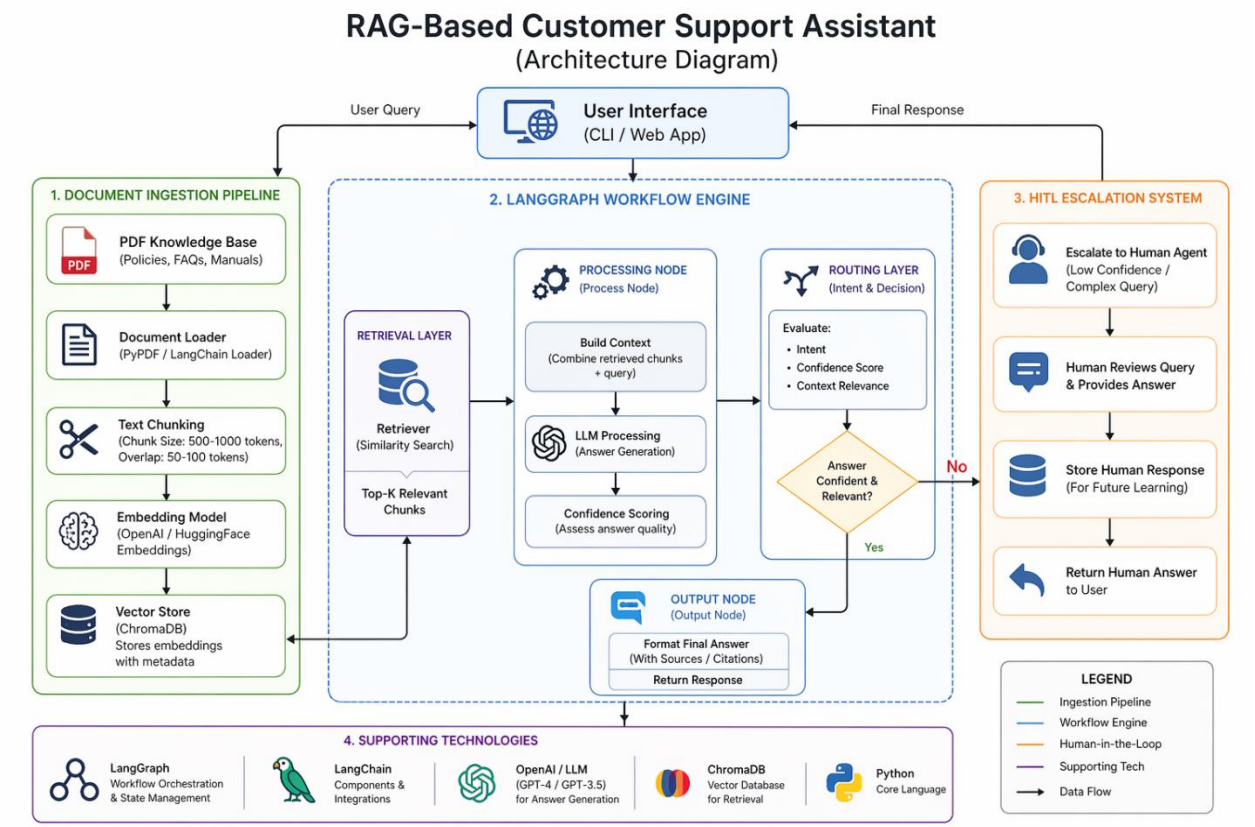
System Overview

This system implements Retrieval-Augmented Generation (RAG) for customer support. It grounds LLM responses in a PDF knowledge base to reduce hallucinations and improve accuracy.

1. Scope: PDF ingestion, semantic retrieval, LLM answer generation, intent-based routing, Human-in-the-Loop (HITL).
2. Non-goals: full CRM integration, real-time streaming updates.

Architecture

Layer	Component	Responsibility
UI	CLI/Web	Accept user queries, display answers
Ingestion	Loader/Chunker/Embedder	Process PDFs into embeddings
Storage	ChromaDB	Persist vectors + metadata
Retrieval	Retriever	Top-k semantic search
LLM	Generator	Answer synthesis with context
Orchestration	LangGraph	Stateful workflow + routing
HITL	Escalation	Human review for low confidence



Data Flow

1 Ingestion: PDF → Text → Chunks → Embeddings → Vector Store
2 Query: User Query → Embed → Retrieve Top-k → Build Context → LLM → Confidence → Route → Output/HITL

Tech Choices (Why)

Choice	Reason
ChromaDB	Lightweight, local dev friendly, fast similarity search
LangGraph	Deterministic, stateful control vs linear chains
OpenAI/HF Embeddings	High semantic quality
GPT-class LLM	Strong reasoning and instruction following

Scalability & Reliability

1. Batch embedding + async ingestion for large PDFs
2. Caching frequent queries; add re-ranking for precision
3. Sharding vector store; background compaction
4. Timeouts, retries, circuit breakers for LLM calls